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understanding, finding and using shapefiles

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What are Shapefiles?

A shapefile (SHP) is a vector data storage format for storing the location, shape, and attributes of geographic features. The format was developed by ESRI and is sometimes referred to as 'an ESRI shapefile'. Shapefiles are the most common GIS file type and are now accepted as an industry-wide format.

The shapefile format defines the geometry and attributes of geographically referenced features in three or more files with specific file extensions that should be stored in the same project workspace.

There are three mandatory files:

Name	Type	
 Lines.shp	Shapefile	
 Points.shp	Shapefile	
 Polygons.shp	Shapefile	
		 Lines.dbf
		 Lines.shp
		 Lines.shx

.shp is a *mandatory* Esri file that gives features their geometry. Every shapefile has its own .shp file that represent **spatial vector data**. For example, it could be points, lines and polygons in a map.

.shx are *mandatory* Esri and AutoCAD shape index position. This type of file is used to search forward and backwards.

.dbf is a standard database file used to store attribute data and object IDs. A .dbf file is *mandatory* for shape files. You can open .DBF files in Microsoft Access or Excel.

For a fuller description of the shapefile format, refer to the ESRI White Paper (July 1998): **ESRI Shapefile Technical Description**

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